EE2015: Electric Circuits and Networks

Tutorial 3 Answer Key
(August 22, 2025)

1. $v_x = 7.5\,\text{V}$

2. (a) $R = 8\,\text{k}\Omega$
   (b) Max power $= 1.152\,\text{W}$

3. Equivalent resistance: $10\,\Omega$
   Short circuit current: $1\,\text{A}$

4. $R_{\text{Th}} = -16.67\,\Omega$, $V_{\text{Th}} = 2\,\text{V}$

5. (a) $1.2857\,\text{A}$
   (b) $8.1\,\text{W}$

6. (a) $V_{\text{Th}}(s) = \dfrac{50}{s}$
   (b) $Z_{\text{Th}}(s) = 2s + 3$
   (c) $v_L(0^+) = 0$, $v_L(\infty) = 31.25\,\text{V}$
   (d) $v_L(t) = 31.25\ (1 - e^{-4t})\ u(t)$